

VIBHAS TALLAPALLI

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EDUCATION

University of Waterloo
BASc in Nanotechnology Engineering

Waterloo, ON
Sep 2025 – Present

TECHNICAL SKILLS

Microcontrollers: Arduino, ARM, AVR, Embedded Systems, Serial Communication, Sensor Integration
Mechanical: SolidWorks, Onshape, CAD, GD&T, Tolerance, DFM/DFA, Carbon Fiber Fabrication, Mill, Lathe, Bandsaw
Software: C++, Python, C, MATLAB, Git, Arduino C++, PID Control, Matplotlib, Tkinter, PPG, IMU

EXPERIENCE

Biomechatronics Engineering Team | *Fusion 360, 3D Printing, Embedded Systems, C, C++*
Mechatronics Prototyping Engineer | *Firmware (Controls/PID)*

Waterloo, ON
Sep 2025 – Present

- Designed thigh/shin strap mount designs for a **lower-body exoskeleton**, building **4+ prototypes** with 3D-printed fixtures to improve comfort, cut slip, and eliminate frame interference while maintaining stability during **walking trials** across 4 contact points over repeated wear tests.
- Integrated **PPG heart-rate sensing**, **step tracking**, vibration alerts, and **GPS/notifications** into the exoskeleton strap for prototypes and final build.
- Built the strap sensor data pipeline (**PPG + IMU**), applying filtering and timestamped logging in **C/C++** to generate clean sessions for **PID** tuning and analysis.

Waterloop Hyperloop | *Onshape, SolidWorks, Machining (Mill/Lathe), Mechanical Assembly*
Mechanical Engineer

Waterloo, ON
Sep 2025 – Jan 2026

- Maintained a **100+ part Onshape assembly** integrating brake suspension battery packaging to coordinate mechanical design across subteams.
- Designed and machined **5 aluminum brake** and pneumatic components from SolidWorks drawings using **mill/lathe workflows**, then installed and fit-checked them on the pod to confirm alignment and clearances, saving **2 weeks of build time**.

Electrium Electric Vehicles | *SolidWorks, Mechanical Packaging, Design for Assembly, Prototyping*
Hardware Design Engineer

Waterloo, ON
Sep 2025 – Jan 2026

- Created a **SolidWorks CAD model** of the **F2025 e-bike** to centralize frame geometry and key dimensions (seat, handlebars, wheels), giving the team a reliable reference for fitment checks and layout decisions while speeding up enclosure/component packaging.
- Redesigned the **battery enclosure redesign** to improve service access and assembly by replacing welded aluminum with a **tempered-glass panel** and revised mounts, reducing welding/hardware requirements and cutting build cost by **\$300+ cost reduction** across 3 prototypes.

IntegraYouth — Executive | *Outreach, Public Speaking, Leadership, Event Planning*
Secretary | *Blog Writer*

Toronto, ON
June 2024 – Present

- Wrote **50+ blog posts** on programs and impact, reaching **1,000+ views** and improving outreach through consistent publishing and topic planning.
- Ran Secretary-team operations and supported event logistics, communications, and scheduling for **5+ events** with **200+ attendees**.

PROJECTS

Adaptive Focus Productivity Timer | *Arduino (C++), Python (Tkinter), Serial Communication, CAD* 

Nov 2025 – Jan 2026

- Developed a **3-sensor embedded occupancy detection system** on **Arduino Uno (C++)**, implementing **10 Hz polling** and cross-checking between sensors to suppress false triggers and maintain stable real-time presence detection.
- Built a **Python Tkinter desktop application** that communicates with Arduino over **serial (9600 baud)** to manage study sessions, log **100+ focus/break events** per session, and visualize performance metrics through a real-time session history dashboard with automatic JSON data logging.
- Designed and fabricated a custom **SolidWorks enclosure** to integrate the **Arduino and sensor array** into a compact, portable unit, improving durability and enabling easy setup across different study environments.

Electric Bike Control System Simulation | *PID Control, C++, Physics Modelling, Python (Matplotlib)* 

Oct 2025 – Nov 2025

- Built a **C++/Python simulation** of an e-bike speed control system, testing **50+ rider load conditions** and analyzing **100+ controller responses** to ensure the bike held a steady target speed while maximizing battery efficiency or feeling unstable when conditions change.
- Implemented and tuned a **PID speed controller** with throttle saturation and anti-windup (integral clamping), reducing **15% overshoot reduction**, **20% settling time improvement**, and improving simulated energy efficiency by 40% under changing load/terrain.

Micro Wind Energy Harvesting Turbine | *CAD (Simulation), Fabrication, Tolerance/GD&T, DC* 

Sept 2025 – Oct 2025

- Designed the **SolidWorks rotor** in SolidWorks to get a tighter, balanced fit on the shaft (less wobble/friction), as well as built it using PVC and nylon through sanding/alignment adjustments, allowing the turbine to spin more smoothly and raising measured **32% DC output voltage increase**.
- Converted rotation into DC output by driving a **DC generator** from the rotor shaft and logging voltage across **10+ wind-test runs** with a multimeter.